

Questions for practice

Q1. Write a Java program to demonstrate the use of a parameterized constructor. Create a `Student` class with the following data members: `rollNo`, `name`, and `marks`. Define a parameterized constructor to initialize these values and display the student details using a member method. Create objects for at least two students and display their details.

Q2. Write a Java program to demonstrate the concept of an interface using a Library Management System. Create an interface `Library` with methods `issueBook()` and `returnBook()`. Implement this interface in a class `LibraryMember`. The program should accept the member name and book name, and display appropriate messages when a book is issued and returned. Create an object of the implementing class and invoke both methods.

Q3. Write a Java program to demonstrate the concept of inheritance using a Library Management System. Create a base class `Book` with data members `bookId`, `title`, and `author`, along with a method to display book details. Create a derived class `EBook` that inherits from `Book` and includes an additional data member `fileSize`. Display all the book details using the derived class object.

Q4. Write a Java program to demonstrate the concept of encapsulation using a Bank Account Management System. Create a class `BankAccount` with private data members `accountNumber`, `accountHolderName`, and `balance`. Provide public getter and setter methods to access and modify these data members. Also, create methods `deposit()` and `withdraw()` to perform banking operations while maintaining proper access control. Create an object of the class and demonstrate the operations.

Q5. Write a Java program to demonstrate different programming paradigms using a simple Online Shopping System. Create a class `Product` with appropriate data members and methods. Demonstrate the use of the following programming paradigms in the program:

1. Object-Oriented Programming
2. Procedural Programming